

Blockchain technology applications and their role in developing the operations and activities of the external trade sector

تطبيقات تقنية سلسلة الكتل ودورها في تطوير عمليات وأنشطة قطاع التجارة الخارجية

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Abstract:

This research deals with the application of Blockchain technology in the field of external trade and the various operations and activities associated with it, from smart contracts, financial operations, insurance, shipping, tracking, administrative and customs procedures, and receipt. This study aims to highlight the solutions provided by this modern and advanced technology to develop the field of international trade and contribute to raising the efficiency of import and export operations and transactions related to the exchange of goods and services between countries and companies across borders and the impact of this on supporting growth and development for countries around the world. We used the descriptive analytical approach. We reached a number of results; the most important of them is the inevitability of adopting this technology with the necessity of unifying international standards and expanding the scope of its use to achieve further development and efficiency in international trade management.

Keywords: Trade; External Trade; International Trade; Digital Technology; Blockchain.

JEL Classification Codes: F10, F13, F30, O33.

ملخص:

يتناول هذا البحث موضوع تطبيق تقنية سلسلة الكتل في ميدان التجارة الخارجية ومختلف العمليات والأنشطة المرتبطة بها من العقود الذكية والعمليات المالية والتأمين إلى الشحن والتتبع والإجراءات الإدارية والجمركية وصولاً إلى الاستلام، حيث تهدف هذه الدراسة إلى إبراز الحلول التي تقدمها هذه التقنية الحديثة والمتطورة لتطوير مجال التجارة الدولية والمساهمة في رفع كفاءة عمليات الاستيراد والتصدير والمعاملات المتعلقة بتبادل السلع والخدمات بين الدول والشركات عبر الحدود وتأثير ذلك على دعم النمو والتنمية للدول عبر العالم. استخدمنا المنهج الوصفي التحليلي، وتوصلنا إلى عدد من النتائج أهمها حتمية تبني هذه التقنية مع ضرورة توحيد المعايير الدولية في هذا المجال وتوسيع نطاق استخدامها لتحقيق مزيد من التطور والكفاءة في إدارة التجارة الدولية.

كلمات مفتاحية: تجارة، تجارة خارجية، تجارة دولية، تكنولوجيا رقمية، سلسلة الكتل.

تصنيفات JEL: F10، F13، F30، O33.

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1. INTRODUCTION

Global trade is one of the main sectors that affect the growth of the global economy. Therefore, developing international trade by enhancing freedom and removing various obstacles that affect commercial activity, such as customs and technical barriers, can increase the productivity of countries and increase the pace of economic growth at the global level.

Blockchain technology is considered one of the most important modern advanced technologies that provide appropriate solutions for developing the field of external trade and international trade due to its characteristics and effective solutions. The use of this technology has grown recently to the point that it has become a sign of a revolution in different fields through which we may enter a new era in which digital currencies and exchanges via databases and digital tools replace traditional models and methods given the degree of security, insurance and reliability that characterizes Blockchain technology, which has begun to take an upward trend gradually across different regions of the world.

Research Problem:

- How can Blockchain applications contribute to the development of external trade? these sub-questions branch out from this problem:
- What are the reasons for countries and companies to apply this technology in developing external trade operations?
- What are the most important operations in which companies apply this technology in various operations in the field of External trade?

Research Hypotheses:

- Blockchain technology provides real, tangible and highly efficient solutions for countries and companies in the field of external trade.
- Blockchain applications in the field of international trade are decentralized digital platforms based on Blockchain and smart contracts, which ensure the provision of multiple services.

Research Objectives: This study aims to:

- Highlight the concept of Blockchain technology, its types, characteristics, and the motives to apply it in external trade.
- Identifying the various applications of Blockchain technology in external trade and the various activities and operations associated with it.

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- Highlighting the economic, social and technical benefits resulting from the application of this technology for the international trade sector, at the level of companies, countries and at the global level as a whole.

2. Blockchain Technology foundations:

2.1. Blockchain Technology: concept, structure, and advantages:

2.1.1 Blockchain Technology definition:

It is a chain of blocks on which information of all types is stored, characterized by transparency, security, and working without a central control body. It is the distributed ledger technology (DLT), which includes digital systems that record asset transactions and their details in multiple locations at the same time. (Belloz & Daoudi, 2024, P.273)

It is a distributed digital ledger that decentralizes the sharing of immutable data and allows information to be stored and transmitted securely without a controlling authority. (LAFORET, 2022, p .52)

Blockchain is a P2P technology that records and verifies transactions in a decentralised, cryptography secure manner. Blockchain applications can globally manage records of any sort such as import-export declarations, invoices, bills of lading (BLs) and Cos. (Macedo, 2018, p. 92)

2.1.2 Blockchain System Elements The Blockchain consists of four main elements: the block, the information, the hash, and the time stamp:

- a. The block** represents the building block of the chain, and is a set of operations or tasks that are desired to be performed or executed within the chain. Examples of blocks include transferring money, recording data, following up on a case, or otherwise. Each block usually accommodates a specific amount of operations and information, and does not accept more than that until the operations within it are finally completed. Then a new block is created associated with it. The main goal is to prevent fictitious transactions within the block that cause the chain to freeze or prevent it from recording and terminating transactions.
- b. Information** refers to the sub-process that takes place within a single block, or it is the "single order" that takes place within the block, and represents, along with other orders and information, the block itself. (Issawi & Doufi, 2020, p. 50)
- c. Hash** is the unique DNA of the block chain, and is sometimes referred to

by some as a "Digital Signature", which is a code produced by an algorithm within the block chain program called the hash function, and it performs four main functions:

- Distinguishing the chain from other chains, as each chain gets its own unique hash.
- Identifying and identifying each block and distinguishing it from others within the chain, as each block also gets its own hash.
- Marking each piece of information within the block with a unique hash.
- Linking blocks to each other within the chain, as each block is linked to the hash that precedes it and the hash that follows it, which makes the hash go in only one direction from the original block that follows it, and so on. It is noted here that the hash does not allow modification to the blocks that have been created.

d. Time stamp: the time of any operation performance within the chain.

2.1.3 Main advantages:

- a. Decentralized:** Blockchain systems work on a central database that is stored as a copy of the general ledger in each device that joined the network, unlike central databases whose storage and management are monopolized by a specific party, which facilitates and accelerates cooperation and work between parties and entities, and weakens the possibility of hacking or losing data and changing or sabotaging it, as it requires hacking more than 50% of the network devices at the same time.
- b. Distribution:** Blockchain offers a distributed ledger, which is a distributed information system that is jointly managed by multiple parties, thus breaking the concept of traditional accounting with the original double entry and becoming a ledger “distributed accounts on a shared network, i.e. accounting Distributed”.
- c. Transparency:** Traditional databases have four classic operations: insert, delete, update and select. Blockchain technology abandons two options, which are deletion and update, leaving only two processes, insertion and selection, through an interconnected structure of blocks, and matching timestamps to link the chain that forms a documented and intertwined set of data that is difficult to tamper with. Thus, blockchain technology enhances the level of transparency in the transaction record compared to

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current record systems, and all changes that occur in the public transaction record book can be seen by all devices joining the network, and are only made with the approval of all relevant parties, which raises the level of transparency and increases trust to a greater extent than in current transaction systems.

- d. Independence:** meaning that each node of the network points is independent of the other and is not affected by it, and is equal to it. Among the advantages of this equality and justice, and among the disadvantages is the difficulty of a central authority controlling it if necessary. (Issawi & Doufi, 2020, p. 51)
- e. Reducing costs and increasing the speed of transactions:** Blockchain systems contribute to reducing costs due to the lack of need for a party An intermediary for completing transactions, as the general transaction log is distributed across all devices joining the network, so any party can enter and settle the transactions that concern him and audit them immediately and directly, which means the speed of completing transactions and avoid additional expenses. (Issawi & Doufi, 2020, p. 52)

2.1.4 Blockchain's strength factors: it's strength lies in two main points:

- a. Decentralization:** The main goal of Blockchain technology is to eliminate the idea of centralization, as there is no single party, server or device that controls the block chain, but rather the chain is distributed among all individuals participating in it around the world, as anyone can download the chain, view it and participate in it.
- b. The inability to manipulate the transaction:** The most prominent feature that distinguishes Blockchain is the absence of fraud or deception during the implementation of transactions conducted through it and the absence of manipulation of transactions after they are completed, which applies to many daily activities such as the transfer of money, shipments and containers, the registration of contracts and properties and the shipping of goods, which helps in creating trust among users to a great extent. (Bou El-Nah & Mouka, 2022, p. 995)

2.2 Blockchain technology: development, types, technical specificities:

2.2.1 Blockchain technology emergence and development:

The term Blockchain first appeared in a paper titled "Published in 2008 by a person or group of people under the name Satoshi Nakamoto (Nakkaz & al, 2021, p. 220), the paper titled Bitcoin: A Peer-to-Peer" "Electronic Cash System (Issawi & Doufi, 2020, p 49), who has remained anonymous until now, and no one knows whether he is a real natural person or a legal entity, it is likely that it is the name of a group of people, while the idea of the blockchain was proposed by Chaum David in 1982, and this technology was actually used initially in the financial field with the emergence of Bitcoin and cryptocurrencies that allowed exchange operations to be carried out on the basis of a completely decentralized model. (Allak, 2023, p. 158)

The Blockchain database has developed in recent decades in line with the development in the field of information technology and levels of artificial intelligence. Since its launch in 2009, Bitcoin was considered the first generation (Blockchain 1.0) of this platform as a cryptocurrency operating on a reliable linear system represented by a general ledger for all transactions. A series of developments followed. By 2013, a second generation of Blockchain (Blockchain 2.0) appeared. It became known as "Ethereum" by a Russian programmer named "Vitalik Butenin" as an update to the public Blockchain as a distributed computing platform using the smart contract feature that ensures sequence and accuracy in organizing transactions in a secure manner, confirming the general terms and conditions of the first generation of this technology. It was updated in 2015, where it attracted developers and technicians from the largest companies in the world of technology. Over time, the scope of the distributed cloud based on the Blockchain database was expanded to a more general platform in which many decentralized applications are included in various fields such as financial and industrial services to become A huge, global cloud in which millions of transactions take place in a short period of time, governed by smart contracts, is known as the third generation killer (Blockchain 3.0) (Kaddouri & Belhabib 2023, p. 151)

2.2.2 Types of Blockchain technology:

Three types of blockchain technology can be distinguished as follows:

- a. Public Blockchain:** it is available to anyone, as it is unusual for a public chain to hide the identity of all participants associated with it. The

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famous cryptocurrency Bitcoin is the most famous example, which has many advantages such as the ability to resist hacking or control capital from central systems. As for its disadvantages, it has high technical requirements in terms of many devices with high capabilities to save this network, and it does not take into account privacy because it is public by nature. (Bou El-Nah & Mouka, 2022, p. 994)

- b. Private Blockchain:** No one can access this chain, only users who are then enabled by administrators, as the private chain is secured by user rights and password. The only difference between it and the public blockchain is that it allows the feature of entering the network through a central figure through rules, conditions and instructions agreed upon by the founder or group of founders.
- c. Dual Blockchain:** It is a combination of public and private Blockchains through which cooperation, coordination and privacy are achieved primarily. (Bou El-Nah & Mouka, 2022, p. 995)

Table 1: Type of Blockchain platforms

Blockchain types	Level of centralization	Read	Write	Example
Public permissionless	Highly decentralized	Anyone	Anyone	Bitcoin
Public permissioned	Highly decentralized	Anyone	Authorized participants	Sovrin
Consortium permissioned	Partially-decentralized	Authorized participants	Authorized participants	Tradelens
Private permissioned	Centralized	Authorized participants	Authorized participants	Company blockchain

Source: (Game, 2021, p. 130)

2.2.3 The technical specificities of Blockchain:

Blockchain comes in response to challenges presented in the current financial system. Indeed, until its creation, the two ways of owning money, that is to say, by cash, or by the bank, display certain flaws:

The possession of tangible currency encourages theft and counterfeiting and prevents its digital transfer to foreign countries.

The traditional banking system has its own challenges. It is indeed possible to own intangible money, but banks then involve transaction fees, storage of easily hackable personal data, and the two billion unbanked people cannot use the money if they do not have a bank account.

The possibility of owning intangible money without the need for an intermediary then becomes possible thanks to Blockchain technology. This revolutionary system appeared in 2008 with the launch of the famous cryptocurrency, bitcoin, introduced by Nakamoto. Cryptocurrency completely disrupts the current banking system, since we move from a "hierarchical, synchronic, vertical" system to a "historical, temporal, diachronic, horizontal dimension" Herlin (2015). Cryptocurrencies, especially bitcoins, can then be done in three different ways: investing a fiat currency (such as the euro) in cryptocurrencies, remuneration by cryptocurrency, which is still not very widespread, and finally by mining cryptocurrencies, explained in the following paragraphs. This system remains without regulation or control. (LAFORET, 2022, p . 54)

2.3 Blockchain technology adoption and spread:

2.3.1 Motivations for Blockchain adoption: The most recurring motivations are summarized in the following elements:

Traceability, allowing companies to provide consumers with information on the life cycle of products and on the implications of companies relating to sustainable development. This information implies an increase in the level of trust on the part of consumers in manufacturers. It is this traceability that represents one of the major applications of blockchain and which appears to be the most popular with companies.

The transparency of data induced by the shared common ledger. It is important in the Collaboration between actors. (LAFORET, 2022, p . 73)

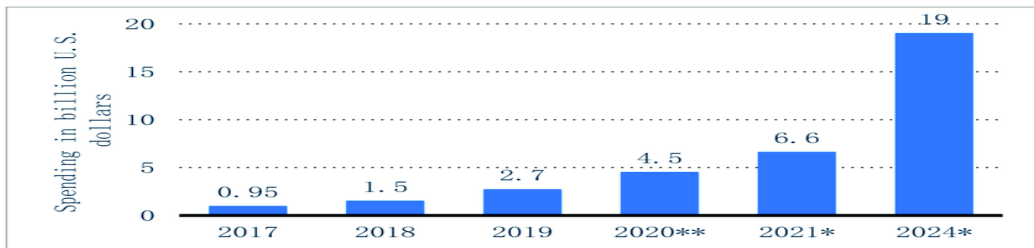
2.3.2 The spread of Blockchain technology: Many countries and companies are working to understand Blockchain technology and explore the various possible applications of this technology in various economic fields and even other fields such as health, education, ...etc (other fields).

The volume of global investment in this technology increased from 0.95 \$ billion in 2017 to 6.6 \$ billion in 2021. It is expected that the volume of investment in Blockchain technology will increase in 2024 to reach 19 \$ billion (Figure1), due to the global trend towards digitizing various economic transactions, due to the changes caused by the Corona pandemic in business models, and to the characteristics of Blockchain technology that allow it to provide solutions to the various problems that different business

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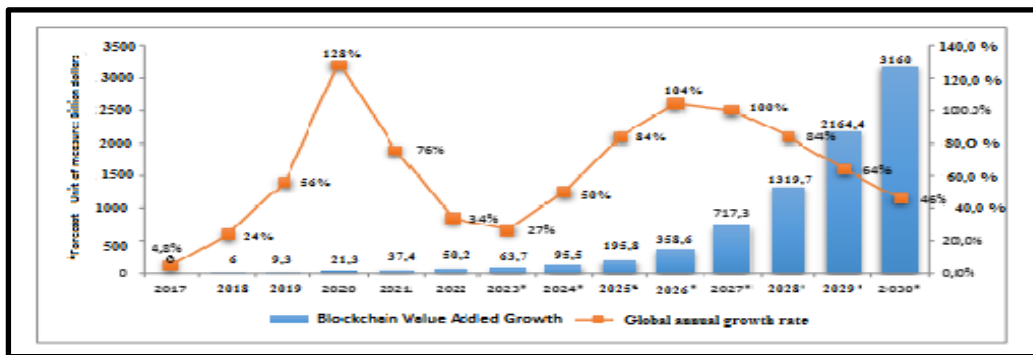
models suffer from. (Kassi & Saadawi, 2022, pp. 113-114)

Figure 1: The volume of global investment in blockchain technology



Source: www.statista.com 2022

**Figure 2 :The reality and expectaions of added value provided by Block
Chain technology to international trade**



Source: www.statista.com 2022

3. International trade: concept, foundatons, and mechanisms:

3.1 The definition of international trade:

3.1.1 Classical approach: The international trade is the process of exchanging goods and services from one place to another, and in a broader sense across borders between different countries and multiple geographical regions in the world, different from the definition of external trade, which is the transfer and movement of goods, services and capital between different countries of the world and what results from or is related to this transfer of operations directly related to this movement such as transportation, insurance and other services that these goods and services need in their movement between different countries. (Abu Al-Oula, 2011, p. 6)

3.1.2 Modern approach: The international trade is the exchange of goods and services between countries; this type of trade contributes to the global economy, as prices, supply and demand affect and is affected by global events. (Hellis, 2022, p.15)

3.2 Motives for international trade and the new world trading system:

3.2.1 Motives for International Trade

- Search for new markets.
- Search for new ideas.
- Get cheap work.
- Get natural materials.
- Get new products. (B. Magee - Translated by: Babiker, 1990, p. 14)

Trade between countries includes two basic types:

- ✓ The first type: when the receiving country cannot produce the goods or provide the services in question itself, or does not have enough.
- ✓ The second type: when the country has the ability to produce the goods or provide the services, but it continues to import them. The reasons for importing this category of products generally fall into three categories: price, the larger volume of goods through imports, the imported goods have other advantages (quality or design). (Sherlock & Reuvid, 2008, p. 03)

3.2.2 The new world trading system is a set of multilateral organizations and mechanisms, the purpose of which is to liberalize international trade.

- a. GATT: Origin and development** The London Conference on Trade and Employment in 1946 resulted in the establishment of a World Trade Organization, which was then called the International Trade Organization. In 1947, the Geneva Conference was held, in which 23 countries participated to negotiate the World Trade Organization (WTO).
- b. World Trade Organization (WTO):** In early 1995, the world witnessed the implementation of the foundations of the new global trade system, as the goal of establishing the World Trade Organization was to create a single institutional framework that includes the GATT Agreement, in addition to all amendments arising from the Uruguay Round, most notably the General Agreement on Trade in Services (GATS), the TRIPS Agreement, and the TRIMS Agreement. (Abdat, 2010, p. 54)

b.1 Objectives of the organization:

- Creating a global competitive position in international trade based on economic efficiency in allocating resources.
- Increasing real income growth rates by maximizing global income and

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raising living standards.

- Increasing the level of both production and global trade and the optimal use and full employment of global resources within the framework of sustainable development.
- Expanding the establishment of new patterns for the division of international labor and increasing global trade scope.
- Involving and integrating developing countries into the new global economic system through their participation in international trade.
- Ensuring that developing countries, especially the least developed, obtain a growing share in global trade in line with their economic development needs. (Abdat, 2010, pp. 55-56)

3.3 External trade liberalization and Globalization of markets:

3.3.1 External trade liberalization programs:

Recent experiences have shown that the success of trade liberalization programs in a country depends on the availability of a stable economic climate and the adoption of sound policies to encourage exports and liberalize imports; therefore, the policies that trade programs must include can be divided into five sections:

- Import policies.
- Export encouragement policies.
- Exchange rate policies.
- Macroeconomic management policies.
- Trade regulatory policies towards trading partners. (Abdat, 2010, p. 56)

3.3.2 Globalization of markets:

The ability to control the non-access no longer exists, so that the world as a whole has become in one cocoon, as all countries now exist in a world characterized by the great ability to move and transfer information, goods, services, individuals, culture, and even viruses. (B. Magee - Translated by: Babiker, 1990, p. 15)

4. Blockchain technology in International Trade:

4. 1Blockchain as alternative to traditional methods and its advantages:

4.1.1 Blockchain technology as alternative to traditional methods:

In international trade Blockchain technology has many advantages compared to traditional methods such as documentary credits and SWIFT

transfers in the field of international trade, as the Blockchain provides transparency, ease of origination and lower costs resulting from reducing intermediaries, in addition to speed in completing transactions thanks to its decentralized nature, and reduces the risk of fraud and facilitates immediate settlements, which makes Blockchain technology traditional. The latter relies on several methods that are more efficient and effective in facilitating international trade operations and exchanging financial information between parties, such as documentary credits issued by the bank as a pledge to pay the beneficiary in exchange for documents that meet specific conditions, and SWIFT transfers that carry out financial transfers between banks via a unified global network, in addition to checks that require time for clearing. These methods rely on financial intermediaries where costs rise and lack the transparency and speed necessary to international trade.

4.1.2 Blockchain in international trade: advantages and opportunities:

- Facilitates international payment operations and money transfers.
- Reduces fraud and protects intellectual property.

In recent years, many pioneering initiatives and projects have emerged based on Blockchain technology in a wide range of sectors. For example, Estonia, the Emirates, and China have used Blockchain in government fields related to real estate records, healthcare, and digital identities. Global companies such as Microsoft, Walmart, FedEx, and Bank of America have adopted Blockchain applications to improve their operations and services, and important financial institutions, including the Bank for International Settlements and the Chinese Central Bank, have issued currencies Digital Blockchain Powered by Blockchain Technology These diverse real-world examples reflect the accelerating adoption and wide scope of Blockchain applications across the world. (Belloz & Daoudi, 2024, P. 274)

4.2 Blockchain Uses in International Trade:

Blockchain technology has penetrated many sectors and has had a major impact on international economic transactions, and smart contracts have become the real solution that provides relative parity to distribute risks equally among participants in international trade

4.2.1 Blockchain in the Shipping and Customs Sector:

Blockchain technology offers advantages that make it useful to the

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shipping and customs sector, including accurately tracking goods and shipments through an immutable record, such as what Singapore did by launching the Singapore Ports Authority platform based on Blockchain technology to track shipments and simplify customs procedures. It also reduces fraud and forgery in documents, such as Maersk, one of the largest shipping companies in the world. Among the companies using Blockchain technology, in 2018, Maersk, in partnership with IBM, launched the TradeLens platform to develop the global supply chain. It allows companies, customs and other stakeholders to access real-time shipping information, which increases the transparency and efficiency. Blockchain technology reduces bureaucratic procedures and lowers shipping costs.

4.2.2 Blockchain in payment settlement:

It is a platform that allows international payments to be settled efficiently and securely, as indicated by a study. It allows cross-border transactions to be carried out quickly and at a low cost, through encrypted distributed ledgers that provide reliable real-time verification of transfers. It also eliminates intermediaries such as correspondent banks, clearing houses and insurance companies. This can significantly reduce costs, time and fees. Blockchain technology adds a high level of transparency and trust to transactions by recording them in a distributed public ledger that all interested parties can access, as indicated, in addition to the decentralized nature of the records system, which increases its continuity. Blockchain represents a promising alternative to traditional payment systems which are more expensive and complex. (Belloz & Daoudi, 2024, P. 275)

4.2.3 Blockchain in the supply chain and logistics sector:

Blockchain has the ability to greatly facilitate supply chain operations globally, and helps in the product path from the manufacturer to the end consumer. Thus, maintaining the quality of these products throughout all stages of the chain, without the need for intermediaries or document authentication or subjection to certain conditions. Rather the process is organized and transparent, which creates trust between the parties.

4.2.4 Blockchain in the banking and financial sector:

Blockchain technology contributes to financing financial activities related to international trade and transactions, including lending, borrowing,

and stock trading, and also contributes to reducing the costs of providing banking services, such as the costs of transferring money, buying and selling assets, etc. (Belloz & Daoudi, 2024, P.273-276)

4.2.5 Smart export based on Blockchain and the Internet of Things:

a. Smart export definition: it the ability to achieve commodity, service, information, financial, cultural and tourism flows to countries and global markets based on digital technology and modern technologies, and thus it is the transit of local goods outside regional borders through customs departments using Blockchain technology and the Internet of Things and automation of the supply chain. (Belloz & Daoudi, 2023, P. 617)

b. Smart Export Steps: a process carried out according to 16 main steps:

- 1) Installing the source system on the Blockchain is recorded in the form of a set of QR codes.
- 2) The exporter negotiates an export deal with the importer.
- 3) The importer places an order for a specific commodity under the code number xxx from the exporter, using the resource discovery framework, where the exporter platform searches for the importers platform for electronic invoicing, and sends the commercial invoice directly to the target platform according to the agreed international standards.
- 4) Since the exporter and importer operate on different platforms, and because the commercial invoice is one of the foundations of trust, the registration of the invoice on the public Blockchain is documented using the notary framework on the ledger through which the importer indicates his acceptance of the invoice is documented.
- 5) Importer creates a review ledger application and stores it on Blockchain.
- 6) The transportation conditions require that products or goods remain at specific temperatures and above certain degrees during shipping, so the exporter uses the services of shippers to ship the goods. Shippers have containers equipped with IoT temperature sensors and GPS tracking.
- 7) Shippers use the smart bill of lading based on the public Blockchain, which replaces paper documents and implements fast, secure, reliable and cost-effective processing of bills of lading. They use the Internet of Things, GPS, banking and financial solutions to provide value-added services such as tracking shipments, location, movement, temperature,

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origin, insurance, quality control and bank guarantee services through smart credit-letter. Cryptocurrencies is preferable (shipping companies).

- 8) The smart bill of lading based on the Blockchain provides users with a superior service at a fraction of the cost, as users of this bill of lading can transfer ownership rights of goods securely and transparently, in seconds.
- 9) Shipping agents are obliged to provide the container ID to the exporter, who is obliged to search and find the container address on the Internet and subscribe to the container data.
- 10) The freight forwarder provides the signed and authenticated invoice and Blockchain reference to the Chamber of Commerce of the exporting country, which in turn verifies the data and issues an automated, signed and Blockchain-registered certificate of origin.
- 11) The freight forwarder creates a shipment reference using its logistics system and provides the shipment ID to the customs of the exporting country to discover the resources and locate the data of this shipment.
- 12) The shipment data includes a reference to the authenticated invoice, container ID, carrier ID and certificate of origin ID, so the customs of the exporting country can discover the complete data about each entity, verify the integrity, and generate an approved export declaration recorded as a smart contract in the inter-enterprise ledger.
- 13) The importer reviews and approves all export and shipping documents and submits the import declaration.
- 14) In the importing country, the customs authority is obliged to print a new import declaration and verify the importer's location of the commercial documents.
- 15) Upon arrival at the importer's port, the container data summary indicates the shipment arrival date, documented temperature, humidity, light, location, door opening, tilt, magnetic field and vibration.
- 16) When the inspection of the product or goods transport platform to the importer's warehouse is completed, the IoT device that supplied the shipment issues a receipt notice. This is in addition to other documented transactions, such as information related to the clearance of the invoice price under the agreed terms. (Belloz & Daoudi, 2023, P. 617- 619)

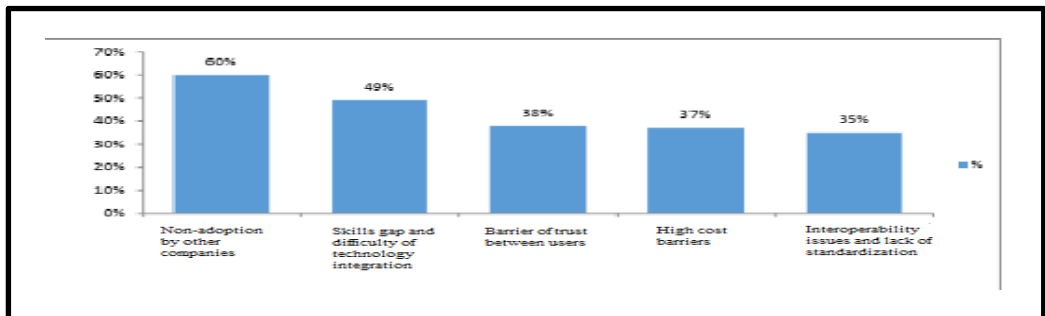
4.2.6 Smart contracts

Smart contracts are software applications that autonomously enforce itself (self-execute) when particular circumstances are satisfied. (based on the if... then..." logic - for example, if goods are dumped at port of X, monies are sent). They specify each party's responsibilities under the "contract," as well as the rewards and fines that may be payable from either party under certain conditions. Unlike a standard legal contract, they may also receive information as input, analyse it according to the contract's rules, and perform any agreed-upon action as a consequence. Such data is given into the smart contract through so-called "oracles," which are data feeds supplied by outside service providers based on pre-defined circumstances. (Kutowaroo, & al, 2023, p. 3759)

4.3 Risks and Obstacles to Applying Blockchain in International Trade

A research study conducted by the Top 5 Productivity Center in 2020, entitled "American Productivity and Quality Center APQC and American Quality Blockchain challenges", where data was collected from companies that use Blockchain technology or are considering adopting it worldwide, in order to identify the challenges and obstacles facing these companies, and after analyzing the data, the results shown in Figure (3) were reached.

Figure 3: Most important risks and obstacles facing Blockchain technology



Source:<https://www.techtarget.com/searchcio/tip/5-challenges-with-blockchain-adoption-and-how-to-avoid-them> (consulted 16/08/2024)

From Figure (3), we note that 60% of the surveyed companies faced the challenge of “non-adoption by other companies,” due to some companies’ lack of confidence in the new technology, while 49% faced the challenge of “using technology” due to lack of technical expertise, 38% faced the challenge of “lack of trust among users” due to the lack of regulatory frameworks, 37% faced the challenge of financial resources due to the high costs of digital transformation, and 35% faced the challenge of the obstacle

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of interoperability or lack of cooperation of separate Blockchains” due to the absence of standards for unifying systems.

- **Lack of adoption by other companies:** both the supplier and the institution are required to participate in this system, the refusal of one of them to participate will lose tracking capabilities in global supply chains.
- **Skills gap and difficulty of technological integration:** The required expertise is difficult to obtain in order to develop and maintain Blockchain and address the technical challenges of integrating it.
- **The barrier of trust between users:** because Blockchain technology is still growing and developing, which raises concerns about data localization, privacy and security issues and determining the applicable law and distributing responsibility for legal recognition of information based on Blockchain. These issues are a barrier to adopt this technology.
- **High cost barriers:** This technology is characterized by high transaction costs, especially for cross-border payments, and the high costs of setting up infrastructure and licensing are added to it, which hinders the adoption of this modern technology, especially for small and medium-sized enterprises.
- **Interoperability issues and lack of standardization standards:** This may create barriers to Blockchain technology adoption and prevent parties from interacting positively. (Belloz & Daoudi, 2024, P.282-283)

5. CONCLUSION

Blockchain technology has become an important tool in various fields and its influence and areas of use are increasing day after day, given the advantages, benefits and solutions it provides to the various fields in which it is applied, which increases efficiency and effectiveness in performance. The external trade sector, with its various operations and related activities, is one of the most important fields that has adopted this technology and has gradually become a pillar and a basic tool in completing international trade transactions. The results achieved encourage the generalization of this technology to include broader dimensions, which contributes to achieving economic results that support global growth, raise the levels of performance of the economy in countries, and increase the profits of companies, which contributes to their development and expansion in production and trade

exchange activities at the international level. Based on the above, we have reached a set of results through this research, which are as follows:

Proving the validity of the main hypotheses, accordingly:

- Blockchain technology provides real, tangible and highly efficient solutions for countries and companies in the field of external trade.
- Blockchain applications in the field of international trade are decentralized digital platforms ensures the provision of multiple services.

We also reached other results, which are:

- Blockchain technology enables a large group of parties to be linked, such as producers, customers, and carriers...etc, which allows the possibility of performing transactions directly between them quickly and securely.
- Blockchain technology provides effective solutions for all parties to international trade operations in a secure manner, which protects all parties from theft or fraud.
- Blockchain technology offered by many specialized companies enables the treatment of various problems that hinder and impede the completion of commercial operations with the least amount of time, effort and cost while ensuring the rights and satisfaction of all parties.
- This technology enables the control of a huge number of transactions reaching millions of operations within one week.
- Blockchain technology has proven its effectiveness and imposed itself as an important pillar for developing international trade.

Research proposals:

- Establishing a unified platform that brings together all parties involved in external and international trade operations.
- Algerian operators must adopt this technology and integrate it into external trade operations to catch up with the development in this field.
- External trade plays a crucial role in economic development; promoting this sector necessitates the adoption of Blockchain technology.

6. Bibliography List:

1. Books :

- 1) Abu Al-Oula, Muhammad Mahmoud (2011): International Trade Systems, University Publishing House, Egypt - Cairo;
- 2) Game, Emmanuelle (2021): Blockchain's Practical and Legal Implications for Global Trade and Global Trade Law (chapter of book entitled from Part II - From E-

Blockchain technology applications and their role in developing the operations and activities of the external trade sector.

Commerce to Big Data and AI), Published online by Cambridge University Press: 09 July 2021, UK; Edited by Mira Burri;

link: <https://doi.org/10.1017/9781108919234.009>

- 3) Hellis, Abdelkader (2022): Advanced Lectures in International Trade Theories, Algeria - University of Djelfa, Faculty of Economics, and Management Sciences;
- 4) Sherlock, Jim & Reuvid, Jonathan (2008): The Handbook of International Trade (2nd Edition), GMB Publishing Ltd, UK – London;
- 5) Stephen, B. Magee - Translated by: Babiker, Hassan Ali (1990): International Trade, Addison-Wesley publishing company, USA-Boston;

2. Theses:

- 6) LAFORET, Liza (2022): Supply Chain & Blockchain Factors of adoption of innovative technology and strategic alignment: case study (doctorate dissertation), UNIVERSITY OF PAU AND THE ADOUR REGION, France.

3. Journal articles :

- 7) Abdat, Murad (2010): International exchange between the proposals of international trade theories and the reality of global economic variables, Journal of the Institute of Economic Sciences, Algeria, University of Algiers 3, Volume 14 - Issue 1, pp. 37– 65; Link: <https://www.asjp.cerist.dz/en/downArticle/258/14/1/70953>
- 8) Allak, Hanan (2023): Using Blockchain Technology to Improve Supply Chain Performance, Shuaa Journal of Economic Studies, Algeria - University of Tissemsilt, Volume 7 - Issue 2, pp. 157-176; Link: <https://www.asjp.cerist.dz/en/downArticle/530/7/2/234492>
- 9) Belloz, Mohamed and Daoudi, Abdel Fattah (2023): Smart Export Using Blockchain Technology and Internet of Things, Journal of Economic and Financial Research, Algeria - University of Oum El Bouaghi, Volume 10 - Issue 1, pp. 608-628; Link: <https://www.asjp.cerist.dz/en/downArticle/64/10/1/228959>
- 10) Belloz, Mohamed and Daoudi, Abdel Fattah (2024): The role of Blockchain technology in developing international trade during the period 2017-2030, Journal of Finance, Investment and Sustainable Development, Algeria - University of Setif 1, Volume 9 - Issue 1, pp. 269-289; Link: <https://www.asjp.cerist.dz/en/downArticle/631/9/1/251197>
- 11) Bou El-Nah, Rima & Mouka, Abdelkrim (2022): Blockchain Technology and Its Applications in External Trade, Journal of Legal and Social Sciences, Algeria - University of Djelfa, Volume 7 - Issue 2, pp. 991 – 1005; Link: <https://www.asjp.cerist.dz/en/downArticle/434/7/2/189372>
- 12) Issawi, Siham & Doufi, Karmia (2020): The development of commercial services using Blockchain technology, Journal of International Economics and Globalization, Algeria - University of Djelfa, Volume 3 - Issue 3, pp. 48 - 60; Link: <https://www.asjp.cerist.dz/en/downArticle/258/14/1/70953>
- 13) Kutowaroo, Ashveen & Dr Jugurnath, Bhavish & Bissessur, Roucheet (2023): Blockchain and International Trade: An Empirical Analysis, Tuijin Jishu/ journal of

propulsion technology, Mauritius- University of Mauritius, volume 44 – Issue 4, pp. 3755-3767;

Link:

<https://www.propulsiontechjournal.com/index.php/journal/article/download/1533/1067/2637>

- 14) Macedo, Leonardo (2018): Blockchain for trade facilitation: Ethereum, eWTP, COs and regulatory issues, World Customs Journal, Volume 12- Issue 2, pp. 87-94;
Link: <https://doi.org/10.55596/001c.116023>
- 15) Nakkaz, Ahmed & Issaoui, Siham & Zahwani, Reda (2021): A prospective study on the impact of using the blockchain in developing the infrastructure of financial markets, Journal of Economic Issue Studies, Algeria - University of Laghouat, Volume 12- Issue 1, pp. 217-234;
Link: <https://www.asjp.cerist.dz/en/downArticle/593/12/1/143747>
- 16) Kaddouri, Mohamed and Abdelkamel, Belhabib (2023): The role of Blockchain technology orientation in the development and upgrading of e-commerce, Journal of Economics and Sustainable Development, Algeria - University of El Oued, Volume 6 - Issue 2, pp. 148-167;
Link: <https://www.asjp.cerist.dz/en/downArticle/598/6/2/231951>
- 17) Kassi, Salem and Saadawi, Mousa (2022): Application of Blockchain Technology in Foreign Trade Logistics (Tradelens Platform as a Model), Journal of Finance, Investment and Sustainable Development, Algeria - University of Setif 1, Volume 7 - Issue02, pp.110-129; Link: <https://www.asjp.cerist.dz/en/downArticle/631/7/2/212218>

4. Internet websites:

- 18) <https://www.asjp.cerist.dz> (consulted 16/08/2024)
- 19) <https://www.researchgate.net> (consulted 16/08/2024)